

Feb 26 2026

$$V(x) = x^T P x$$

$$P = I: V(x) = x^T x$$

$$\begin{aligned}\dot{V}(x) &= \dot{x}^T x + x^T \dot{x} \\ &= x^T (A^T + A) x\end{aligned}$$

may be sign-definite even if A is Hurwitz

(see example on next page)

Ex. $A = \begin{bmatrix} -1 & 4 \\ 0 & -1 \end{bmatrix}$

$\lambda = -1, -1$ Hurwitz

$$A^T = \begin{bmatrix} -1 & 0 \\ 4 & -1 \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} -2 & 4 \\ 4 & -2 \end{bmatrix} = 2 \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$\begin{array}{ll} \text{trace} < 0 & \Leftrightarrow \sum_i \lambda_i < 0 \\ \det < 0 & \Leftrightarrow \prod_i \lambda_i < 0 \end{array} \left. \vphantom{\begin{array}{l} \text{trace} \\ \det \end{array}} \right\} \text{one } \lambda \text{ is positive}$$

$A + A^T$ is not Hurwitz

$\cdot Q = I$
 $\downarrow$
 P
 $\downarrow$
estimate K



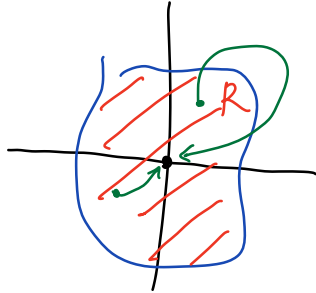
Region of Attraction

$$\dot{x} = f(x)$$

$$x=0 \in \mathbb{R}^n$$

$$R = \{x \mid \phi(t, x) \rightarrow 0\}$$

$\phi(t, x) \equiv x(t)$ starting from x at $t=0$



R is invariant:

$$\phi(t, x) \rightarrow 0$$

$$x \in R \Rightarrow x(s) \in R \quad \forall s$$

$$\Rightarrow \phi(t, x(s)) \xrightarrow{?} 0$$

$$\Rightarrow \phi(t, \phi(s, x)) \xrightarrow{?} 0$$

$$\Rightarrow \phi(t+s, x) \xrightarrow{?} 0$$

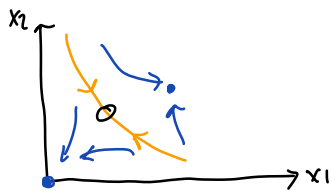
Ex. $\dot{x}_1 = -a x_1 + b x_2$

$$\dot{x}_2 = \frac{x_1^2}{1+x_1^2} - b x_2$$

$$x_2 = a x_1$$

$$\frac{x_1^2}{1+x_1^2} = ab x_1$$

$$\Rightarrow x_1 = ab(1+x_1^2)$$



R is an open, connected, invariant set

Boundaries of R are trajectories

In practice, use $V(x)$ (Lyapunov function) for under-approx of R :

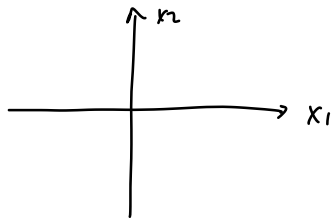
$\Rightarrow$ find largest c such $\Omega_c \subset R$

$$\uparrow$$
$$\dot{V}(x) < 0$$

Ex. $\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix} x$

$$\dot{x}_1 = -2x_1$$

$$\dot{x}_2 = 0$$



When A is diagonalizable

$$x(t) = e^{\lambda_1 t} \langle \omega_1^*, x(0) \rangle v_1$$

$$+ \dots + e^{\lambda_n t} \langle \omega_n^*, x(0) \rangle v_n$$

Laprunov's Linearization Method

$$\dot{x} = f(x) = Ax + \underbrace{(f(x) - Ax)}_{g(x)}$$

$$A = \frac{\partial f}{\partial x} \Big|_{x=0}$$

$$\dot{x} = Ax$$

$$g(0) = 0$$

$$\exists \delta > 0, r > 0 \text{ s.t.}$$

$$V(x) = x^T P x$$

$$|x| < r$$

$$A^T P + P A = -Q$$

$$\Rightarrow |g(x)| < \delta |x|$$

$$\dot{V}(x) = \dot{x}^T P x + x^T P \dot{x}$$

$$= x^T P (Ax + g(x)) + (Ax + g(x))^T P x$$

$$= x^T (PA + A^T P) x + 2x^T P g(x)$$

$$= -x^T Q x + 2x^T P g(x)$$

$$|Ax| \leq \|A\| |x|$$

$$\leq -x^T Q x + 2|x| \|P\| |g(x)|$$

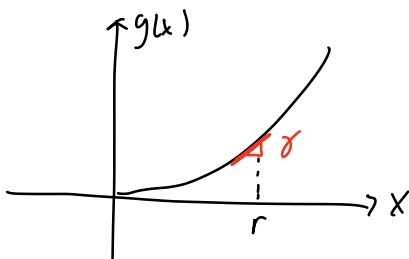
$$\lambda_{\min}(Q) |x|^2 \leq x^T Q x \leq \lambda_{\max}(Q) |x|^2$$

$$\dot{V}(x) \leq -\lambda_{\min}(Q) |x|^2 + 2|x| \|P\| |g(x)|$$

$$\leq -\lambda_{\min}(Q) |x|^2 + 2\delta |x|^2 \|P\|$$

$$\delta < \frac{\lambda_{\min}(Q)}{2\|P\|}$$

$$\Rightarrow \dot{V}(x) < 0$$



Ex.

$$\dot{x} = -\frac{x}{1+t}$$

$$x(t) = x(t_0) \frac{1+t_0}{1+t}$$

$$\dot{x}(t) = A(t) x$$

$$t_0 > -1$$

$$\dot{x} = f(x, t)$$